Financial Ratios: Academic Notes

1. Gross Profit Margin

Gross Profit Margin is a financial metric that indicates the proportion of revenue that exceeds the cost of goods sold (COGS). It reflects how efficiently a firm produces or sources its products. The formula to calculate Gross Profit Margin is:

Gross Profit Margin = [(Revenue - COGS) / Revenue] x 100

This ratio is particularly useful for understanding a company's core profitability, excluding indirect costs such as administrative and marketing expenses. A high gross margin implies a firm retains a significant portion of revenue as gross profit, suggesting operational efficiency. However, gross margins vary widely across industries, necessitating peer comparison for accurate evaluation.

2. Net Profit Margin

Net Profit Margin measures the percentage of revenue that remains as net income after all operating expenses, interest, and taxes have been deducted. The formula is:

Net Profit Margin = (Net Profit / Revenue) x 100

This metric offers insight into the overall profitability of an enterprise and reflects how effectively all business activities, including cost management, financing, and taxation, are controlled. Higher net margins signify financial strength, while declining margins could suggest inefficiencies or cost pressures.

3. EBIT (Earnings Before Interest and Taxes)

EBIT represents the profit a firm generates from its core operations, excluding the effects of capital structure (interest) and tax policies. It is computed as:

EBIT = Revenue - COGS - Operating Expenses

or

EBIT = Net Income + Interest + Taxes

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As an indicator of operational performance, EBIT is a critical metric for assessing a firm's ability to generate profits from core business activities, independent of its debt obligations and tax environment.

4. EBITDA (Earnings Before Interest, Taxes, Depreciation, and Amortization)

EBITDA expands upon EBIT by excluding non-cash expenses such as depreciation and amortization. It is calculated as:

EBITDA = EBIT + Depreciation + Amortization

This metric is widely used to evaluate a company's cash-generating capabilities and is especially relevant for firms with substantial fixed assets or those in capital-intensive industries. It facilitates inter-company comparisons by neutralizing the effects of varying capital structures and accounting methods.

5. Liquidity Ratios

Liquidity ratios assess a firm's ability to meet its short-term obligations. Key ratios include:

a) Current Ratio = Current Assets / Current Liabilities

- Ideal benchmark: around 2:1
- Indicates general liquidity; higher values suggest better short-term financial health.

b) Quick Ratio = (Current Assets - Inventories) / Current Liabilities

- A more stringent test of liquidity by excluding less liquid inventory.
- Ideal benchmark: >= 1

These ratios are vital for assessing whether a business can maintain operations without raising external capital in the short term.

6. Capital Structure Ratios

These ratios evaluate the composition of a company's long-term financing through debt and equity. Key indicators include:

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a) Debt-to-Equity Ratio = Total Debt / Total Equity

- Measures the degree of financial leverage.
- A high ratio implies greater financial risk, though it may enhance return on equity.

b) Equity Ratio = Total Equity / Total Assets

- Reflects the proportion of assets financed by shareholders.
- Higher ratios indicate financial stability and lower reliance on debt.

c) Interest Coverage Ratio = EBIT / Interest Expense

- Gauges a firm's ability to cover interest payments from operating profits.
- Values above 2.5 are generally considered safe; values below 1 may indicate solvency issues.

Together, these ratios provide insight into the sustainability of a firm's capital structure and its exposure to financial risk.